Jennifer Grannen

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OBJECTIVE

My goal is to build intelligent, adaptable robots that collaborate with humans and other robots to perform complex manipulations and learn new behaviors on the fly.

EDUCATION

Stanford University, Stanford, CA 2021 –
Ph.D. in Computer Science, Artificial Intelligence
Advisor: Dorsa Sadigh

University of California, Berkeley, Berkeley, CA 2021
B.A. in Computer Science GPA: 3.9/4.0
High Distinction in General Scholarship

EXPERIENCE

Stanford Artificial Intelligence Lab, Stanford, CA. 2021–
Ph.D. Student, *Advised by Dorsa Sadigh*
Developing intelligent robotic systems that learn to collaborate between two robot arms or between robots and humans to perform dexterous manipulations.

Toyota Research Institute, Los Altos, CA. 2024
Research Intern, *Advised by Blake Wulfe, Russ Tedrake*
Building interactive robots capable of grounded planning and dexterous movements by leveraging real-time human feedback and vision-language model priors.

Toyota Research Institute, Cambridge, MA. 2021
Research Intern, *Advised by Eric Cousineau, Russ Tedrake*
Designed a system for robotic manipulation of 3D deformable objects, including dough.

Berkeley Artificial Intelligence Research, Berkeley, CA. 2019–2021
Undergraduate Researcher, *Advised by Ken Goldberg*
Created interpretable representations for robotic manipulation of 1D and 2D deformable objects, including knot untangling and clothes folding.

Berkeley RISELab (now Sky Computing Lab), Berkeley, CA. 2019–2020
Undergraduate Researcher, *Advised by Joey Gonzalez*
Developed algorithms to improve language model calibration for sentiment analysis.

Goldman Sachs, New York, NY. 2019
Securities Strategy Summer Analyst
Added features to support new products and risk measures on front facing AI-based application to decrease quote latency.

PUBLICATIONS

[9] **Jennifer Grannen***, Sidd Karamcheti*, Suvir Mirchandani, Percy Liang, Dorsa Sadigh. Vocal Sandbox: Continual Learning and Adaptation for Situated Human-Robot Collaboration. *Conference on Robot Learning (CoRL), 2024.* **Long Oral Presentation (30/693=4.3%)**. [website]

[8] **Jennifer Grannen**, Yilin Wu, Brandon Vu, Dorsa Sadigh. Stabilize to Act: Learning to Coordinate for Bimanual Manipulation. *Conference on Robot Learning (CoRL), 2023.* **Long Oral Presentation (33/498=6.6%)**. [website]

[7] Lorenzo Shaikewitz*, Yilin Wu*, Suneel Belkhale*, **Jennifer Grannen**, Priya Sundaresan, Dorsa Sadigh. In-Mouth Robotic Bite Transfer with Visual and Haptic Sensing. *International Conference on Robotics and Automation (ICRA), 2023.* [website]

[6] **Jennifer Grannen***, Yilin Wu*, Suneel Belkhale, Dorsa Sadigh. Learning Bimanual Scooping Policies for Food Acquisition. *Conference on Robot Learning (CoRL), 2022.* [website]

[5] Vainavi Viswanath*, **Jennifer Grannen***, Priya Sundaresan*, Brijen Thananjeyan, Ashwin Balakrishna, Ellen Novoseller, Jeffrey Ichnowski, Michael Laskey, Joseph E. Gonzalez, Ken Goldberg. Disentangling Dense Multi-Cable Knots. *International Conference on Intelligent Robots and Systems (IROS), 2021.* [website]

[4] Priya Sundaresan*, **Jennifer Grannen***, Brijen Thananjeyan, Ashwin Balakrishna, Jeffrey Ichnowski, Ellen Novoseller, Minho Hwang, Michael Laskey, Joseph E. Gonzalez, Ken Goldberg. Untangling Dense Non-Planar Knots by Learning Manipulation Features and Recovery Policies. *Robotics: Science and Systems (RSS), 2021.* [website]

[3] Aditya Ganapathi, Priya Sundaresan, Brijen Thananjeyan, Ashwin Balakrishna, Daniel Seita, **Jennifer Grannen**, Minho Hwang, Ryan Hoque, Joseph E. Gonzalez, Nawid Jamali, Katsu Yamane, Soshi Iba, Ken Goldberg. Learning Dense Visual Correspondences in Simulation to Smooth and Fold Real Fabrics. *International Conference on Robotics and Automation (ICRA), 2021.* [website]

[2] **Jennifer Grannen***, Priya Sundaresan*, Brijen Thananjeyan, Jeffrey Ichnowski, Ashwin Balakrishna, Minho Hwang, Vainavi Viswanath, Michael Laskey, Joseph E. Gonzalez, Ken Goldberg. Untangling Dense Knots by Learning Task-Relevant Keypoints. *Conference on Robot Learning (CoRL), 2020.* **Long Oral Presentation (20/485=4.1%)**. [website]

[1] Priya Sundaresan, **Jennifer Grannen**, Brijen Thananjeyan, Ashwin Balakrishna, Michael Laskey, Kevin Stone, Joseph E. Gonzalez, Ken Goldberg. Learning Rope Manipulation Policies Using Dense Object Descriptors Trained on Synthetic Depth Data. *International Conference on Robotics and Automation (ICRA), 2020.* [website]

AWARDS

National Science Foundation Graduate Research Fellowship	2021
UC Berkeley EECS Honors Degree	2021
Timothy B. Campbell Innovation Award, UC Berkeley EECS	2021
Upsilon Pi Epsilon CS Honors Society, UC Berkeley	2019
Tau Beta Pi Engineering Honors Society, UC Berkeley	2018

PROFESSIONAL SERVICE

Reviewing.
Conference on Robot Learning (CoRL): 2024.
International Journal of Robotics Research: 2024.
Stanford AI PhD Admissions Reader: 2023.
Robotics: Science and Systems (RSS): 2023, 2024.
Transactions on Robotics (T-RO): 2023
International Conference on Robotics and Automation (ICRA): 2021, 2022, 2023, 2025.
Robotics and Automation Letters (RA-L): 2022, 2023.
International Conference on Automation Science and Engineering (CASE): 2022.
Neural Information Processing Systems (NeurIPS): 2021.

Mentoring. I have been fortunate to work with the following undergraduate and Master's students:

Brandon Vu, 2023–
Yvonne Obukehwo, 2023–2023
Nik Sridhar, 2023–2023
Yilin Wu, 2021–2023 (next, PhD Student at CMU)
Jeff He, 2022–2023
Kaushik Shivakumar, 2021–2021 (next, Software Engineer at Google Deepmind)
Vainavi Viswanath, 2020–2021 (next, Machine Learning Engineer at Google)

TEACHING & OUTREACH

Stanford Computer Science Department, Course Assistant 2024
CS 221: Artificial Intelligence: Principles and Techniques

Stanford CS Mentorship Program, Organizer 2024 –
Pairing undergraduate students with graduate student mentors and organizing related socials with a focus on increasing the participation of underrepresented minorities in CS.

Stanford LINXS, Mentor 2023
Advised an undergraduate student for eight weeks to perform computer vision and robotics research.

Stanford CS Mentorship Program, Mentor 2022
Mentored undergraduate students about future career, opportunities within research labs, and applying to industry jobs or graduate school.

Stanford AI4ALL, Instructor 2022
Developed curriculum for and lead a three week course to teach computer vision concepts to high school students, culminating in a project using neural networks to sort trash.

UC Berkeley EECS Department, Teaching Assistant 2020–2021
CS 188: Introduction to Artificial Intelligence

UC Berkeley AUTOLAB, Outreach Lead 2020–2021
Organized workshops to present robot demos to visiting middle and high school students.

UC Berkeley Society of Women Engineers, Instructor 2019–2021
Lead weekly classes teaching middle school girls Scratch and Python.

Expanding Your Horizons, UC Berkeley, Undergraduate Mentor 2017–2020
Organized workshops and science experiment activities for an annual intro to STEM conference for middle school girls.